

11.12

The price of a non-dividend paying stock is \$19 and the price of a three-month European call option on the stock with a strike price of \$20 is \$1. The risk-free rate is 4% per annum. What is the price of a three-month European put option with a strike price of \$20?

According to put-call parity:

$$c + D + Ke^{(-rT)} = p + S_0$$

$$1 + 0 + 20e^{(-0.04 \cdot \frac{3}{12})} = p + 19$$

$$p = 1 + 20e^{-0.01} - 19$$

$$p = 20 \cdot 0.99004983374 - 18 = 19.8009966748 - 18 = 1.8009966748 \approx \boxed{\$1.80}$$

11.19

The price of a European call that expires in six months and has a strike price of \$30 is \$2. The underlying stock price is \$29, and a dividend of \$0.50 is expected in two months and again in five months. Interest rates (all maturities) are 10%. What is the price of a European put option that expires in six months and has a strike price of \$30?

$$c + D + Ke^{(-rT)} = p + S_0$$

$$2 + 0.50(e^{-0.10 \cdot \frac{2}{12}} + e^{-0.10 \cdot \frac{5}{12}}) + 30e^{(-0.10 \cdot \frac{6}{12})} = p + 29$$

$$p = 2 + 0.50(0.98347145382 + 0.95918945711) + 30e^{-0.05} - 29$$

$$p = 2 + 0.97133045546 + 28.536882735 - 29$$

$$p = 2.50821319046 \approx \boxed{\$2.51}$$

11.20

Explain the arbitrage opportunities in Problem 11.19 if the European put price is \$3.

We found in Problem 11.19 that the fair put price should be \$2.51, so if this price is actually \$3, then there is an arbitrage opportunity with the put being overpriced and the call being underpriced. We can therefore **short both the put and underlying stock** and then also **buy the call**. We first immediately make $-2 + 3 + 29 = \$30$, and we would then profit in the corresponding scenarios for the final stock price:

- If the stock price is above \$30, we exercise the call option and buy the stock for \$30, or with PV $30e^{-0.10 \cdot \frac{6}{12}} = 30 \cdot 0.9512294245 = 28.536882735 \approx 28.54$, and the dividends on the short cost $0.50(e^{-0.10 \cdot \frac{2}{12}} + e^{-0.10 \cdot \frac{5}{12}}) = 0.50(0.98347145382 + 0.95918945711) = 0.97133045546 \approx 0.97$, so we end up with a PV-adjusted profit of $30 - 28.54 - 0.97 = \$0.49$, which is exactly the difference in the put price.
- If the stock price is below \$30, the put we shorted is exercised, so we are still forced to buy the stock for \$30, with the same PV calculation of before as 28.54. We still are short the stock, so we pay the same 0.97 dividend as before. The profit is therefore the same \$0.49 as before.

11.23

Prove the result in equation (11.7). (Hint: For the first part of the relationship consider (a) a portfolio consisting of a European call plus an amount of cash equal to K and (b) a portfolio consisting of an American put option plus one share.)

11.24

Prove the result in equation (11.11). (Hint: For the first part of the relationship consider (a) a portfolio consisting of a European call plus an amount of cash equal to $D + K$ and (b) a portfolio consisting of an American put option plus one share.)

11.26

Use the software DerivaGem to verify that Figures 11.1 and 11.2 are correct.

11.31

Suppose that c_1, c_2 , and c_3 are the prices of European call options with strike prices K_1, K_2 , and K_3 , respectively, where $K_3 > K_2 > K_1$ and $K_3 - K_2 = K_2 - K_1$. All options have the same maturity. Show that

$$c_2 \leq 0.5(c_1 + c_3)$$

(Hint: Consider a portfolio that is long one option with strike price K_1 , long one option with strike price K_3 , and short two options with strike price K_2 .)

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Consider an option on a stock when the stock price is \$41, the strike price of \$40, the risk-free rate is 6%, the volatility is 35%, and the time to maturity is 1 year. Assume that a dividend of \$0.50 is expected after six months.

(a)

Use DerivaGem to value the option assuming it is a European call.

(b)

Use DerivaGem to value the option assuming it is a European put.

(c)

Verify that put-call parity holds.

(d)

Explore using DerivaGem what happens to the price of the options as the time to maturity becomes very large and there are no dividends. Explain your results.